Question	Answer	Marks	AO	Guidance
6	**DR**			
	$\tan^2 x - 3 = 2\tan x$	**M1^**	**3.1a**	Multiplying through by $\tan x$
	$\tan^2 x - 2\tan x - 3 = 0$	**M1^**	**1.1**	Get all 3 terms of quadratic on one side and zero on the other
	$(\tan x - 3)(\tan x + 1) = 0$ $\tan x = 3,\ \tan x = -1$	**M1***	**1.1**	Solve 3 term quadratic to get both values of $\tan x$ **Must** see either use of the formula, factorisation or completing the square. Allow 1 error in method Condone missing '= 0'
	71.6°, 251.6°, 135°, 315°	**A1^** **A1***	**1.1** **1.1**	**A1^** for any two roots correct (dep. on **M2^**) **A1*** for all roots with no additional solutions (dep. on **M1***) Accept 72 and 252
	Alternative solution			
	$\frac{\sin x}{\cos x} - 3\frac{\cos x}{\sin x} = 2$	**M1^**	**3.1a**	Use of $\tan x = \frac{\sin x}{\cos x}$
	$\sin^2 x - 2\sin x\cos x - 3\cos^2 x = 0$	**M1^**	**1.1**	Get all 3 terms on one side and zero on the other
	$(\sin x - 3\cos x)(\sin x + \cos x) = 0$ $\tan x = 3,\ \tan x = -1$	**M1***	**1.1**	Factorise to get both values of $\tan x$ **Must** see factorisation. Allow 1 error in method Condone missing =0
	71.6°, 251.6°, 135°, 315°	**A1^** **A1***	**1.1** **1.1**	**A1^** for any two roots correct (dep. on **M2^**) **A1*** for all roots with no additional solutions (dep. on **M1***) Accept 72 and 252
		[5]		

Q6. This is another DR question. There are 2 solutions provided.
The first method involves multiplying through by tan x (M1) and the second method involves replacing tan x with sin x/cos x and cot x with cos x/sin x (M1). Both methods then involve getting an equation with 3 terms on one side and zero on the other (M1) and solving their 3-term equation AND get both values of tan x (M1). A correctly solved equation implies any missing M marks. They **must** solve the equation by a valid method (factorisation, formula or completing the square) to get the M1 here.

The A marks are for giving the angles – A1 for any two correct (subject to at least M2) and A2 for all the roots with no additional solutions (subject to M3).

It might be quite common to give M1 M1 M0 A1 A0 because they fail to solve the equation and just write down answers. Note we are allowing 72°/ 252° and also over specified answers. e.g. 71.565°/ 251.565°.